Experiment No. 5

Title: Write a program to find the longest common subsequence between two sequences.

Aim: The aim of this program is to find the longest common subsequence (LCS) between two given sequences. The LCS of two sequences is the longest sequence that can be derived from both original sequences by deleting some or no elements without changing the order of the remaining elements. This problem has applications in bioinformatics, text comparison, and various other fields.

Theory: What is Dynamic Programming (DP)?

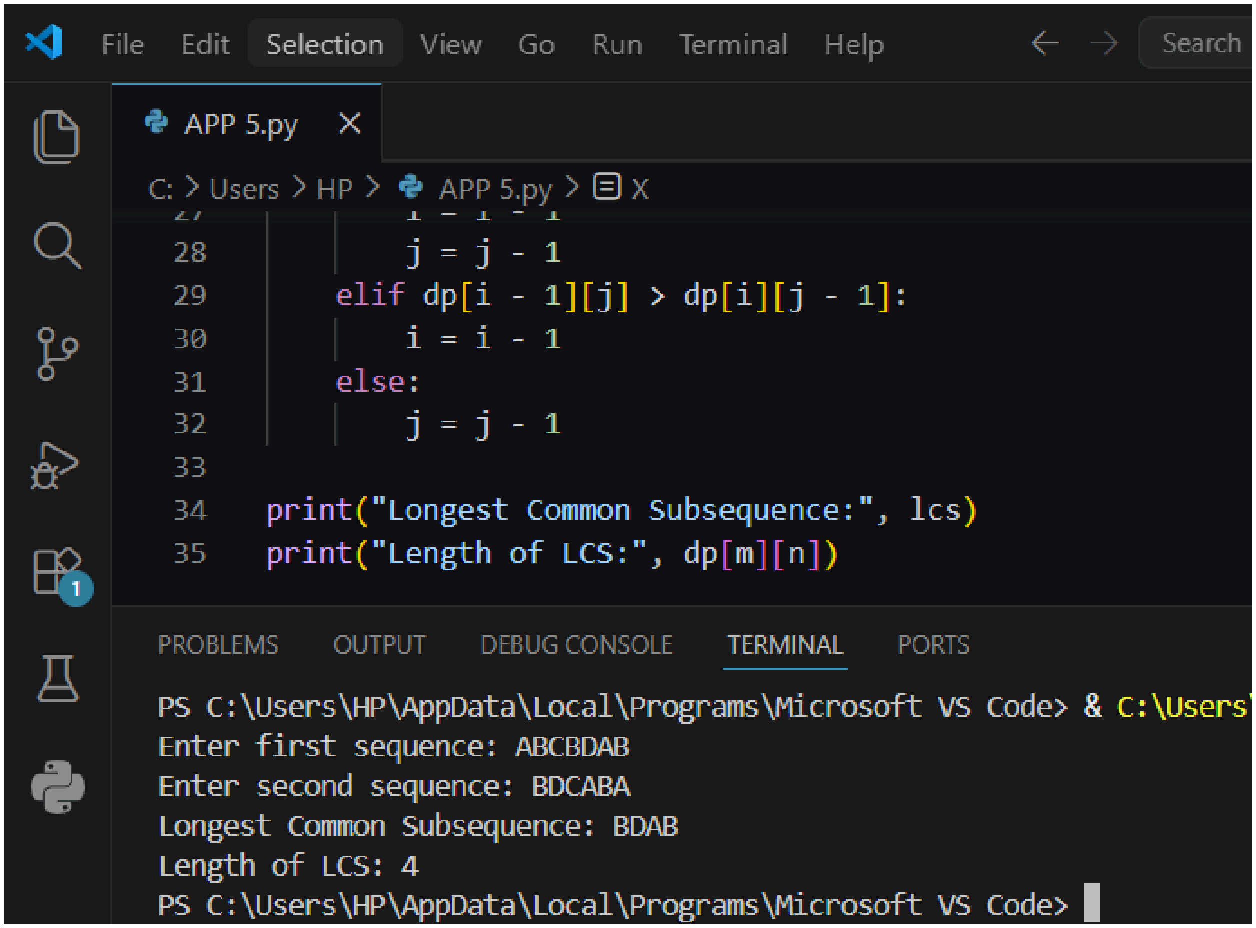
Dynamic Programming (DP) is a method used in mathematics and computer science to solve complex problems by breaking them down into simpler subproblems. By solving each subproblem only once and storing the results, it avoids redundant computations, leading to more efficient solutions for a wide range of problems.

Experiment 5 Code:

```
X = input("Enter first sequence: ")
Y = input("Enter second sequence: ")

m = len(X)
n = len(Y)

dp = [[0 for j in range(n + 1)] for i in range(m + 1)]

for i in range(1, m + 1):
    for j in range(1, n + 1):
        if X[i - 1] == Y[j - 1]:
            dp[i][j] = dp[i - 1][j - 1] + 1
        else:
            if dp[i - 1][j] > dp[i][j - 1]:
                dp[i][j] = dp[i - 1][j]
            else:
                dp[i][j] = dp[i][j - 1]

i = m
j = n
lcs = ""

while i > 0 and j > 0:
    if X[i - 1] == Y[j - 1]:
        lcs = X[i - 1] + lcs
        i = i - 1
        j = j - 1
    elif dp[i - 1][j] > dp[i][j - 1]:
        i = i - 1
    else:
        j = j - 1
```

```
print("Longest Common Subsequence:", lcs)
print("Length of LCS:", dp[m][n])
```

Experiment 5 Output (Screenshot):



Conclusion:

Hence, we studied to determine the longest common subsequence between two sequences by employing a dynamic programming approach, which efficiently computes the optimal subsequence length through iterative table-filling.